

AI 3603 ARTIFICIAL INTELLIGENCE: PRINCIPLES AND TECHNIQUES

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I. CODING PART

In this section, we implemented the core functions for exact inference in Bayesian Networks using the Variable Elimination algorithm.

A. Implementation

1. *joinFactors(Factor1, Factor2)*

This function computes the product of two factors.

- We first identify the common variables between the two factors.
- If common variables exist, we use `pd.merge` to perform an inner join on these variables.
- If no common variables exist, we perform a cross join (Cartesian product) to combine every row of `Factor1` with every row of `Factor2`.
- The new probability for each row is calculated as the product of the probabilities from the two input factors: $P_{new} = P_1 \times P_2$.
- Finally, the redundant probability columns are dropped.

2. *marginalizeFactor(factorTable, hiddenVar)*

This function sums out a variable from a factor (marginalization).

- We identify all variables in the factor table except the `hiddenVar` and the probability column.
- We use `pd.groupby` on these remaining variables and sum the probabilities using `.sum()`.
- This effectively computes $\sum_{hiddenVar} P(\dots)$, resulting in a factor that no longer depends on `hiddenVar`.

3. *evidenceUpdateNet(bayesnet, evidenceVars, evidenceVals)*

This function reduces the factors based on observed evidence.

- We iterate through each factor in the Bayesian network.
- For each evidence variable present in a factor, we filter the DataFrame to keep only the rows where the variable matches the observed value.
- This operation corresponds to selecting the slice of the probability distribution consistent with the evidence.

4. *inference(bayesnet, hiddenVars, evidenceVars, evidenceVals)*

This function performs the Variable Elimination algorithm to compute the posterior distribution $P(Query|Evidence)$.

- We call `evidenceUpdateNet` to restrict all factors to the observed evidence values.

- For each variable in `hiddenVars`:
 1. We identify all factors that contain the current hidden variable.
 2. We join these factors into a single combined factor using `joinFactors`.
 3. We marginalize out the hidden variable from this combined factor using `marginalizeFactor`.
 4. The resulting factor is added back to the pool of factors, replacing the original ones.
- After eliminating all hidden variables, we join all remaining factors to form the final joint distribution.
- We normalize the resulting probabilities so that they sum to 1, ensuring a valid probability distribution.

B. Results

The implementation was tested using `BayesNetworkTestScript.py`. The results correspond to the examples in Bishop’s *Pattern Recognition and Machine Learning* (Chapter 8) and the RiskFactor dataset.

1. Bishop Textbook Examples

Marginal Probability of Gauge ($P(\text{Gauge})$)

Gauge	Probability
0	0.315
1	0.685

TABLE I: Inference result for $P(\text{Gauge})$

Conditional Probability of Gauge given Fuel=0 ($P(\text{Gauge}|\text{Fuel} = 0)$)

Fuel	Gauge	Probability
0	0	0.81
0	1	0.19

TABLE II: Inference result for $P(\text{Gauge}|\text{Fuel} = 0)$

Conditional Probability of Fuel given Gauge=0 ($P(\text{Fuel}|\text{Gauge} = 0)$)

Fuel	Gauge	Probability
1	0	0.7429
0	0	0.2571

TABLE III: Inference result for $P(\text{Fuel}|\text{Gauge} = 0)$

Conditional Probability of Fuel given Gauge=0, Battery=0 ($P(\text{Fuel}|\text{Gauge} = 0, \text{Battery} = 0)$)

Battery	Fuel	Gauge	Probability
0	1	0	0.8889
0	0	0	0.1111

TABLE IV: Inference result for $P(\text{Fuel}|\text{Gauge} = 0, \text{Battery} = 0)$

2. RiskFactor Data Test

Inference for $P(\text{Diabetes}|\text{Smoke} = 1, \text{Exercise} = 2, \text{Long_sit} = 1)$

Smoke	Long_sit	Exercise	Diabetes	Probability
1	1	2	1	0.1363
1	1	2	2	0.0091
1	1	2	3	0.8377
1	1	2	4	0.0169

TABLE V: Inference result for Diabetes risk given specific lifestyle factors.

II. WRITTEN PART

In this section, we analyzed risk factors for certain health problems (heart disease, stroke, heart attack, diabetes). The data is from the 2015 Behavioral Risk Factor Surveillance System (BRFSS) survey shown in FIG. 1.

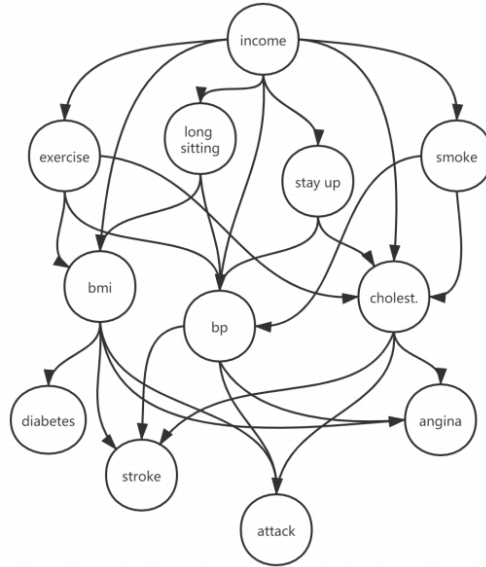


FIG. 1: BRFSS

Task 1: Network Construction

We defined the network structure as a dictionary mapping each child variable to its list of parent variables. For each node, we invoked `readFactorTablefromData` to compute its Conditional Probability Table (CPT) by counting the occurrences of each child-parent configuration in the `RiskFactorsData.csv` dataset. We then summed the number of entries in all CPTs to determine the Bayesian network size. The full joint distribution size was calculated as the product of the domain sizes of all variables.

Quantity	Value
Bayesian network size (sum of CPT entries)	1048
Full joint distribution size (product of domain sizes)	131072

Task 2: Health Outcome Queries

We performed inference using the variable elimination algorithm implemented in `inference()`. For each query, we specified the target outcome variable, the evidence variables (habits or health status), and their observed values. The function automatically identified the remaining variables as hidden variables and marginalized them out.

- For **2(a)**, we set evidence for **Bad Habits** (smoke=1, exercise=2, long_sit=1, stay_up=1) and **Good Habits** (smoke=2, exercise=1, long_sit=2, stay_up=2).
- For **2(b)**, we set evidence for **Poor Health** (bp=1, cholesterol=1, bmi=3) and **Good Health** (bp=3, cholesterol=2, bmi=2).

We considered four outcomes: **diabetes**, **stroke**, **attack**, **angina**.

Task 2(a) Probability of outcome = 1 under Bad vs Good habits

TABLE VI: $P(\text{outcome} = 1 \mid \text{habits})$

Outcome	Bad Habits	Good Habits
Diabetes	0.179685	0.074406
Stroke	0.053783	0.029390
Attack	0.085606	0.036247
Angina	0.095094	0.034808

Task 2(b) Probability of outcome = 1 under Poor vs Good health

TABLE VII: $P(\text{outcome} = 1 \mid \text{health status})$

Outcome	Poor Health	Good Health
Diabetes	0.114420	0.056981
Stroke	0.083736	0.014633
Attack	0.141073	0.015994
Angina	0.161065	0.013123

Task 3: Effect of Income

We iterated through income levels from 1 to 8. In each iteration, we set `income` as the evidence variable with the current level value. We then ran the `inference()` function for each of the four health outcomes to compute the posterior probability $P(\text{outcome} = 1 \mid \text{income} = i)$. The results were collected and plotted to visualize the trend.

We computed $P(\text{Outcome} = 1 \mid \text{income} = i)$ for income levels $i = 1, \dots, 8$.

From [VIII](#) and [FIG. 2](#), we can observe a general trend where higher income levels are associated with lower probabilities of health issues. Specifically, as income increases from level 1 to 8, the risk for diabetes, stroke, heart attack, and angina consistently decreases. This suggests a negative correlation between income and these health risks, likely due to better access to healthcare, healthier lifestyle choices, and reduced stress associated with higher income.

TABLE VIII: $P(\text{outcome} = 1 \mid \text{income level})$

Income	Diabetes	Stroke	Attack	Angina
1	0.147373	0.049657	0.073073	0.078695
2	0.143179	0.051489	0.076085	0.082153
3	0.140931	0.048877	0.071938	0.077174
4	0.137719	0.046780	0.068823	0.073491
5	0.135240	0.044515	0.065479	0.069554
6	0.135321	0.041685	0.061607	0.065119
7	0.133164	0.037980	0.056053	0.058672
8	0.122438	0.033439	0.048949	0.050373

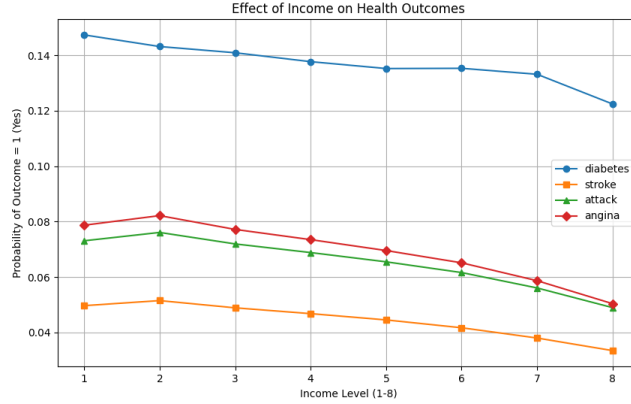


FIG. 2: Effect of Income on Health Outcomes. The plot shows the probability of each outcome (diabetes, stroke, attack, angina) decreasing as income level increases from 1 to 8.

Task 4: Test Assumptions (adding habit $\rightarrow$ outcome edges)

Assumption: The absence of direct links between habits and outcomes in the original graph (Task 1) implied that the effects of these habits on health outcomes were **fully mediated** by the intermediate physiological variables (BMI, blood pressure, cholesterol). This assumed that smoking and exercise did not influence disease risk through any other pathways other than changing a person’s BMI, BP, or cholesterol levels.

We created a new network structure by modifying the parent lists for the outcome variables. Specifically, we added **smoke** and **exercise** to the parent list of **diabetes**, **stroke**, **attack**, and **angina**. We then re-estimated the CPTs for these modified nodes using `readFactorTablefromData` on the original dataset. Finally, we repeated the queries from Task 2(a) and 2(b) using this updated network to observe the impact of direct habit-outcome dependencies.

Task 4(a) $P(\text{outcome} = 1 \mid \text{Bad vs Good habits})$ TABLE IX: $P(\text{outcome} = 1 \mid \text{habits})$

Outcome	Bad Habits	Good Habits
Diabetes	0.246691	0.055762
Stroke	0.080882	0.019590
Attack	0.135632	0.021058
Angina	0.138041	0.023596

Task 4(b) $P(\text{outcome} = 1 \mid \text{Poor vs Good health})$

TABLE X: $P(\text{outcome} = 1 \mid \text{health status})$

Outcome	Poor Health	Good Health
Diabetes	0.120134	0.055176
Stroke	0.083740	0.014719
Attack	0.140240	0.016014
Angina	0.160417	0.013126

Effect: The results showed that adding these direct edges generally increased the estimated probability of adverse outcomes for individuals with "Bad Habits" compared to the original network (Task 1). This suggests that the original model might have underestimated the risk associated with poor lifestyle choices by forcing all influence to pass through the measured intermediates.

Validity: The assumptions of the first graph (Task 1) were **not fully valid**. While BMI, BP, and cholesterol are major mediators, they do not capture the *entirety* of the biological impact of smoking and exercise. For instance, smoking causes direct damage to vascular endothelium and increases clotting risk independent of cholesterol levels. Therefore, the first graph was an oversimplification that likely missed significant direct pathways of disease causation.

Task 5: Interaction (adding diabetes $\rightarrow$ stroke edges)

Assumption: The absence of edges between health outcomes in the earlier networks implied that these outcomes were **conditionally independent** given their parents (risk factors). This assumed that having one condition did not directly influence the risk of developing another, except through shared underlying causes like BMI, blood pressure, or cholesterol.

Starting from the Net 2 structure (Task 4), we further modified the network by adding **diabetes** to the parent list of **stroke**. We re-estimated the CPT for **stroke** to capture the conditional dependence $P(\text{stroke} \mid \text{bmi}, \text{bp}, \text{cholesterol}, \text{smoke}, \text{exercise}, \text{diabetes})$. We then computed the posterior probability of stroke given diabetes status in both the Net 2 (Task 4) and Net 3 (Task 5) networks to compare the results.

TABLE XI: $P(\text{stroke} = 1 \mid \text{diabetes})$ – Q4 (no diabetes $\rightarrow$ stroke) vs Q5 (with diabetes $\rightarrow$ stroke)

Condition	Network Q4	Network Q5
diabetes=1	0.044863	0.077491
diabetes=3	0.040323	0.034832

Effect: Adding the edge **diabetes** $\rightarrow$ **stroke** significantly changed the conditional probability of stroke. Specifically, $P(\text{stroke} = 1 \mid \text{diabetes} = 1)$ increased from $\approx 4.5\%$ (Net 4) to $\approx 7.7\%$ (Net 5), while the probability for non-diabetics ($\text{diabetes} = 3$) decreased slightly. This indicated that the model without the edge underestimated the stroke risk for diabetic patients.

Validity: The assumption of no interaction was **invalid**. The substantial change in probability upon adding the edge suggests a strong direct correlation (or unmeasured common cause) between diabetes and stroke that was not fully explained by the shared parents (BMI, BP, cholesterol). Biologically, diabetes is a known independent risk factor for vascular diseases like stroke, confirming that the interaction model (Net 5) is more valid than the independence model.